Question 1

(a) $L = (\partial_x + x\partial_y)$

$$\begin{aligned} L(u+v) &= (\partial_x + x\partial_y)(u+v) \\ &= u_x + v_x + xu_y + xv_y \\ &= Lu + Lv \\ L(cu) &= (\partial_x + x\partial_y)(cu) \\ &= cu_x + cxu_y \\ &= c(u_x + xu_y) \\ &= cLu \end{aligned}$$

Therefore L is a linear operator.

(b) $L = (\partial_x + u_y\partial_y)$

$$\begin{aligned} L(u+v) &= (\partial_x + (u_y+v_y)\partial_y)(u+v) \\ &= u_x + v_x + (u_y+v_y)u_y + (u_y+v_y)v_y \\ &= u_x + v_x + u_y^2 + 2u_yv_y + v_y^2 \\ &\neq u_x + u_y^2 + v_x + v_y^2 = Lu + Lv \end{aligned}$$

Therefore, L is NOT a linear operator.

(c) $L = (\partial_x + \partial_y + \frac{1}{u})$

$$\begin{aligned} L(u+v) &= (\partial_x + \partial_y + \frac{1}{u+v})(u+v) \\ &= u_x + v_x + u_y + v_y + 1 \\ &\neq (u_x + u_y + 1) + (v_x + v_y + 1) = Lu + Lv \end{aligned}$$

Therefore, L is NOT a linear operator.

(d) $L = (\cos(x)\partial_{xxy} + \arctan(\frac{y}{x})\partial_y)$

$$\begin{aligned}
L(u+v) &= (\cos(x)\partial_{xxy} + \arctan(\frac{y}{x})\partial_y)(u+v) \\
&= \cos(x)u_{xxy} + \arctan(\frac{y}{x})u_y + \cos(x)v_{xxy} + \arctan(\frac{y}{x})v_y \\
&= Lu + Lv \\
L(cu) &= (\cos(x)\partial_{xxy} + \arctan(\frac{y}{x})\partial_y)(cu) \\
&= c\cos(x)u_{xxy} + c\arctan(\frac{y}{x})u_y \\
&= c(\cos(x)u_{xxy} + \arctan(\frac{y}{x})u_y) \\
&= cLu
\end{aligned}$$

Therefore, L is a linear operator.

Question 2

Suppose that v and w are solutions to the inhomogeneous linear PDE $Lu = g$. Then we have

$$Lv = g \qquad\qquad Lw = g$$

Take their difference

$$\begin{aligned}
Lv - Lw &= g - g \\
L(v-w) &= 0
\end{aligned}$$

Since the PDE is linear, $v - w$ is a solution to the homogeneous equation $Lu = 0$.

Question 3

(a)

$$\begin{aligned}
u_x &= 0 \\
u &= f(y)
\end{aligned}$$

(b)

$$\begin{aligned}
u_x &= x^2 + y \\
u &= \frac{1}{3}x^3 + yx + f(y)
\end{aligned}$$

(c)

$$\begin{aligned} u_{yy} + u &= 0 \\ u &= -u_{yy} \\ &= f(x)\cos(y) + g(x)\sin(y) \end{aligned}$$

(d)

$$\begin{aligned} u_{xy} + u_y &= 0 \\ u_y &= -u_{xy} \\ u &= -u_x \\ u &= \cos(x) + f(y) \end{aligned}$$

Question 4

(a)

Question 5

(a)

Question 6